CS2009- DESIGN AND ANALYSIS OF ALGORITHMS
Assignment -2
Fall 2026

Total Marks	Due Date	Late Policy
100	24th October 2026	20% 1-day late, 40% 2-day late No Submissions afterwards

Instructions

- Show all important steps of your solution.
- Zero marks on plagiarism.
- A scanned copy of the handwritten assignment in **PDF format** is required for submission.
- Write and draw using ballpoint only.
- Take your time and focus on writing clearly
- Attempt all questions in the order they are listed here. Otherwise, your submission will not be graded.

Algorithm Design

1. Write an algorithm for finding whether two polygons intersect. Make sure it handles all cases. Provide justification.
2. Given a star-shaped polygon (which has a kernel), describe an algorithm to compute its convex hull. Justify why standard convex hull algorithms may need modification.

3. Is there another method for checking whether a point is inside or outside a polygon??
   If yes specify any two.
4. Given a set of points in a region, describe an algorithm to detect whether they form multiple disjoint convex polygons (clustering + convex hull per cluster).

PROBLEM SOLVING

5. Solve the following problem (unsorted fig.1 and sorted fig.2 version) using the brute force, Quick hull, D&C and Grahams scan and Jarvis march approach. Show all steps. Skipping steps will not give you full marks.

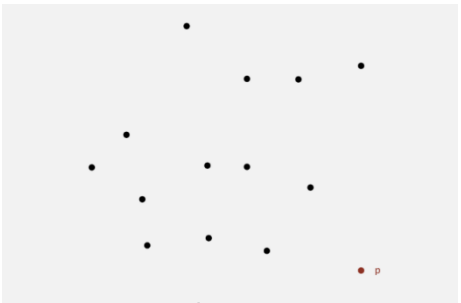


Figure 1

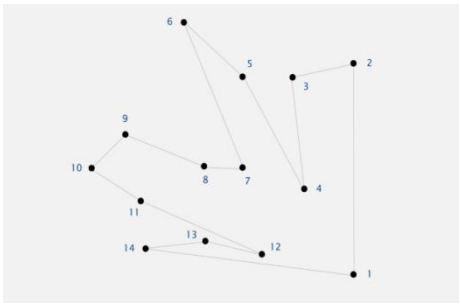


Figure 2

6. A weather management team has deployed 12 weather sensors in stations around a flood-affected region. They need to determine the smallest boundary covering all stations to define the outer operational zone for rescue teams. The weather management team must calculate this boundary using a convex hull algorithm and verify the correctness of the results before the system is deployed. The location coordinates of the weather stations are: $\mathrm{A}\,(-1, 0), \mathrm{B}(1, 3), \mathrm{C}(3, 5), \mathrm{D}(5, 4), \mathrm{E}(7, 1), \mathrm{F}(5, -3), \mathrm{G}(2, -4), \mathrm{H}(-2, -2), \mathrm{I}(2, 1), \mathrm{J}(3, 0), \mathrm{K}(1, 1), \mathrm{L}(4, 2)$ Apply the Quick Hull algorithm to determine the hazard area boundary of the rescue teams. Show the initial partition (line from leftmost to rightmost point), identify points on each side, perform cross-product calculations, select the farthest point, recursively partition, and determine the final hull.

7. A local traffic law-and-order company records vehicle trajectories at a busy intersection. Each vehicle's predicted path is represented by a direction on the road in a $2\mathrm{D}$ coordinate system. The system must

detect whether any two vehicle paths overlap or cross so that potential accidents can be identified. The system receives the following data from the sensors, where each data point represents a vehicle's path between two locations (x, y) points. ((1,1), (5,5)), ((1,5), (5,1)), ((5,5), (5,1)), ((2,2), (4,4)), ((0,0), (0,4))

i. Determine all overlapping and non-overlapping pairs, distinguishing between proper crossing occurring inside both pairs and intersections occurring at endpoints.
ii. Design a brute-force algorithm that examines all possible pairs of locations and reports every crossing pair, and derive its worst-case time complexity for (n) location points.
iii. Assume that the system is expanded to process $100{,}000$ locations. Explain the computational limitations of the brute-force approach and recommend a more efficient computational geometry algorithm, justifying your choice in terms of time complexity and the number of crossings reported.

8. Suppose the dataset grows to $n=100{,}000$ points. Compare the expected worst-case work of: Brute Force convex hull: $O(n^3)$ in a straightforward implementation , Graham Scan: $O(n \log n)$ and Jarvis March: $O(nh)$. Calculate the asymptotic operation counts for $n=100{,}000$ under the assumption that h=100 for Jarvis March. Use the expressions above, not measured execution times. Explain which algorithm you would choose if the data were processed once versus repeatedly as new points arrived. Describe how numerical overflow, duplicate points, and collinear points could cause errors in a real implementation.

9. A cyber security development team is developing a GPS system for an industrial facility whose red zone area is represented by an irregular polygon with vertices $(A(1,1))$, $(B(7,1))$, $(C(9,4))$, $(D(7,8))$, $(E(4,7))$, $(F(2,9))$, and $(G(0,5))$, arranged in counterclockwise order. During system testing, four suspicious attacks are detected at different GPS locations $(P(4,4))$, $(Q(8,6))$, $(R(10,5))$, and $(S(4,1))$. The security system must determine whether each attack is located inside the red zone area, outside it, or exactly on its boundary. Justify the final classification of each attack point. Show all necessary calculations and justify the final classification of each vehicle.

10. A country's defensive system has 12 security IoT sensors installed at different locations. The system must determine the smallest protection boundary that encloses all IoT security sensors to define a secure and defensive region. A (1, 1), B (3, 4), C (5, 7), D (7, 4), E (9, 1), F (8, 6), G (6, 3), H (4, 2), I (2, 5), J (5, 1), K (7, 2), L (5, 5)
    1. Plot all 12 points on a coordinate plane.
    2. Identify the leftmost, rightmost, lowest, and highest points.
    3. Apply the Brute Force Convex Hull algorithm by checking pairs of points and determining whether all remaining points lie on the same side of each line.
    4. Calculate the convex hull using the orientation formula.
    5. Apply Graham Scan and show the sorted points, stack operations, and final hull.
    6. Compare the results of Brute Force, Graham Scan and Quick Hull.
    7. State the time complexity of each algorithm.
    8. Compute the polygon's area using the shoelace formula:

$$A = \frac{1}{2}\left|\sum_{i=1}^{h}(x_i y_{i+1} - y_i x_{i+1})\right|$$

Important: Points lying on a hull edge must be handled carefully. A point can be on the boundary without being a necessary vertex of the minimal polygon.